

THE TWENTY-SIXTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION

November 20, 1965

- A-1. Let ABC be a triangle with angle $A < \text{angle } C < 90^\circ < \text{angle } B$. Consider the bisectors of the external angles at A and B , each measured from the vertex to the opposite side (extended). Suppose both of these line-segments are equal to AB . Compute the angle A .
- A-2. Show that, for any positive integer n ,

$$\sum_{r=0}^{[(n-1)/2]} \left\{ \frac{n-2r}{n} \binom{n}{r} \right\}^2 = \frac{1}{n} \binom{2n-2}{n-1},$$

where $[x]$ means the greatest integer not exceeding x , and $\binom{n}{r}$ is the binomial coefficient “ n choose r ,” with the convention $\binom{n}{0} = 1$.

- A-3. Show that, for any sequence $a_1, a_2, \dots$ of real numbers, the two conditions

(A)
$$\lim_{n \rightarrow \infty} \frac{e^{ia_1} + e^{ia_2} + \dots + e^{ia_n}}{n} = \alpha$$

and

(B)
$$\lim_{n \rightarrow \infty} \frac{e^{ia_1} + e^{ia_4} + \dots + e^{ia_{n^2}}}{n^2} = \alpha$$

are equivalent.

- A-4. At a party, assume that no boy dances with every girl but each girl dances with at least one boy. Prove that there are two couples gb and $g'b'$ which dance whereas b does not dance with g' nor does g dance with b' .
- A-5. In how many ways can the integers from 1 to n be ordered subject to the condition that, except for the first integer on the left, every integer differs by 1 from some integer to the left of it?
- A-6. In the plane with orthogonal Cartesian coordinates x and y , prove that the line whose equation is $ux+vy=1$ will be tangent to the curve $x^m+y^m=1$ (where $m>1$) if and only if $u^n+v^n=1$ and $m^{-1}+n^{-1}=1$.

- B-1. Evaluate

$$\lim_{n \rightarrow \infty} \int_0^1 \int_0^1 \dots \int_0^1 \cos^2 \left\{ \frac{\pi}{2n} (x_1 + x_2 + \dots + x_n) \right\} dx_1 dx_2 \dots dx_n.$$

- B-2. In a round-robin tournament with n players $P_1, P_2, \dots, P_n$ (where $n>1$), each player plays one game with each of the other players and the rules are such that no ties can occur. Let w_r and l_r be the number of games won and lost, respectively, by P_r . Show that

$$\sum_{r=1}^n w_r^2 = \sum_{r=1}^n l_r^2.$$

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- B-3. Prove that there are exactly three right-angled triangles whose sides are integers while the area is numerically equal to twice the perimeter.
- B-4. Consider the function

$$f(x, n) = \frac{\binom{n}{0} + \binom{n}{2}x + \binom{n}{4}x^2 + \cdots}{\binom{n}{1} + \binom{n}{3}x + \binom{n}{5}x^2 + \cdots},$$

- where n is a positive integer. Express $f(x, n+1)$ rationally in terms of $f(x, n)$ and x . Hence, or otherwise, evaluate $\lim_{n \rightarrow \infty} f(x, n)$ for suitable fixed values of x . (The symbols $\binom{n}{r}$ represent the binomial coefficients.)
- B-5. Consider collections of unordered pairs of V different objects $a, b, c, \dots, k$. Three pairs such as bc, ca, ab are said to form a triangle. Prove that, if $4E \leq V^2$, it is possible to choose E pairs so that no triangle is formed.
- B-6. If A, B, C, D are four distinct points such that every circle through A and B intersects (or coincides with) every circle through C and D , prove that the four points are either collinear (all of one line) or concyclic (all on one circle).